

Building a Nigerian Telecommunications Network

Detailed Market-Entry Research with MTN and Airtel as Case Studies
Prepared for Samuel · 29 September 2026 · 2-3 Year Horizon

1. Executive Summary

This research answers three questions: what it takes to build a mobile telecommunications network in Nigeria in the class of MTN and Airtel, what it costs, and the realistic path for a new entrant with a 2-3 year horizon. The headline finding: a full national Mobile Network Operator (MNO) requires roughly **\$600 million to over \$1 billion** in committed capital. However, the Nigerian Communications Commission (NCC) has opened a structured, affordable entry door through its 2023 Mobile Virtual Network Operator (MVNO) licensing framework, with license fees from N30 million to N250 million. The recommended strategy is a staged entry: begin as a facility-based MVNO, prove the brand and subscriber base, add fibre infrastructure, then graduate toward full operator status via spectrum acquisition from existing holders.

2. Case Study: MTN Nigeria

- **Entry (2001):** MTN and Econet (now Airtel) each paid **\$285 million** for one of Nigeria's first digital mobile licenses, at the famous auction run by the NCC.
- **5G Spectrum (2021):** MTN paid **\$273.6 million** for 100 MHz of 3.5 GHz 5G spectrum, alongside Mafab Communications (same price). Airtel joined the 5G race later at comparable cost.
- **Scale today:** roughly **80-100 million subscribers** (Nigeria is MTN Group's largest market), annual revenue around **N3.36 trillion** (2024 audited), approximately **18,000-19,000 base stations**, covering 2G, 3G, 4G and 5G.
- **Annual capex:** around **N440 billion per year** on core infrastructure (fibre, base stations, technology upgrades); cumulative investment in Nigeria exceeds \$10 billion since 2001.

Lesson: MTN's moat is not the license alone. It is two decades of compounding infrastructure, distribution (hundreds of thousands of agents), brand trust, and a tower/network economy of scale that cannot be replicated quickly by capital alone.

3. Case Study: Airtel Nigeria

- **Entry lineage:** entered as Econet (2001, \$285m license), then Vmobile, Celtel, Zain, Bharti Airtel — each transition reflecting consolidation economics.
- **Scale today:** approximately **17,200 base stations** (1,561 new sites deployed in the last three years), roughly **60 million subscribers**, Nigeria being the largest single market of Airtel Africa.
- **Financials:** Nigeria generates around 30% of Airtel Africa group revenue; group capex has been rising yearly, with Nigeria taking a large share of tower and 4G expansion.
- **Strategy note:** Airtel's growth playbook is capacity-led — new sites, spectrum refarming to 4G, and mobile money (smartcash) riding the network.

Lesson: Airtel proves the second-mover advantage works in Nigeria if you out-invest on sites and distribute aggressively. It also proves the market profitably supports at least two large operators.

4. What a Network Actually Costs

Cost Item	Indicative Amount (USD)	Notes
NCC Mobile/Unified license	included in spectrum era auctions; New national mobile UASL yes or no, full service capex	9mobile (2008)
3.5GHz 5G spectrum (100MHz)	\$273.6 million per lot	2021 auction price paid by MTN and Mafab
Digital mobile license (2001 precedent)	\$285 million	Historical benchmark
One base station / tower	\$150,000 - \$200,000	Before energy, backhaul and rent
National coverage (2,000-5,000 sites)	\$300m - \$1 billion+	MTN/Airtel class coverage needs thousands of sites
Core network, IT, billing, fibre backhaul	\$80m - \$200m	EPC/5G core, OSS/BSS, data centres
Annual operating capex (MTN benchmark)	\$140 billion/yr	Maintenance + growth spending

Total realistic project size for a true national network from scratch: **\$600 million to \$1 billion+** (over a trillion naira). This is the MTN/Airtel door, and no new player has walked through it since 2008.

5. The Three Doors Into Nigerian Telecom Today

Door 1 — Full MNO (MTN/Airtel class)

Requires a Unified Access Service License (UASL) from the NCC plus spectrum (auction or transfer). Capital wall: \$600m+. No new national entrant in nearly two decades. Verdict: not the entry door for a 2-3 year plan, but a possible Year 5+ destination.

Door 2 — MVNO (the smart door)

The NCC's 2023 MVNO framework lets you ride a host network's towers (MTN, Airtel, Glo, 9mobile) while owning your brand, SIMs, tariffs, and customers. This is exactly how Lebara Nigeria and Interswitch entered. License tiers: **Tier 1 N30m; Tier 2 N65m; Tier 3 N100m; Tier 4 N150m; Tier 5 (facility-based, own core network) N250m** — with market reports of Tier 5 licences changing hands at up to N500m (Interswitch, May 2023). Tenure: 10 years. **Important:** the NCC has since temporarily suspended new MVNO licence issuance — preparation now positions you to move the moment it reopens, or to acquire an already-licensed operator.

Door 3 — FNO / fibre and data infrastructure

Fixed Network Operator and ISP-class licenses are cheaper and let you own real infrastructure city by city. Fibre assets strengthen any future MNO/MVNO bid and generate revenue early.

6. Regulatory Requirements (All Doors)

- Company registration with the CAC; telecom share capital benchmarks are high (foreign participation guidance around N100 million+).
- NCC licence (individual licence framework): technical and financial capability review; note the new NCC/CAC rule requiring prior NCC consent for share transfers above 10%.
- Equipment type approval from the NCC for all radio/network equipment.
- Interconnection agreements with existing operators; numbering resources from the NCC.
- SIM registration and NIN-SIM linkage integration; QoS obligations and annual operating levies (a percentage of revenue) to the NCC.
- Consumer protection and data protection compliance (NCC regulations + NDPR/NDPA).

7. Recommended 2-3 Year Path

Year 1 — Position

Register the company; engage an NCC licensing consultant; target an MVNO Tier 3-4 licence (N100m-N150m) when issuance reopens, or open acquisition talks with an existing MVNO licence holder. Design the brand, SIM product, tariff and distribution model in parallel.

Years 1-2 — Launch

Sign a wholesale (host network) agreement; launch branded SIMs and data bundles. Realistic working capital on top of the licence: **N1-N2 billion** for distribution, marketing, and operations (Interswitch spent N500m on licence entry alone). Target: prove subscriber growth and revenue.

Years 2-3 — Deepen and bridge

Add fixed fibre in one city (FNO licence) to own infrastructure and margins. Watch the secondary spectrum market — Mafab's attempted sale of its 5G spectrum to MTN proves spectrum can be acquired from existing holders. With proven revenue, raise outside capital (N100bn+) toward a full operator position.

8. Hard Truths

- Capital alone does not beat MTN and Airtel at their game. New entrants win on product, brand, distribution and experience — the Jio-of-India path: product-led entry first, infrastructure later.
- Energy and forex are the two killers of Nigerian networks: every tower runs on power you pay for, and much network cost is dollar-denominated.
- The licence is the easy part. The recurring economics of thousands of powered sites is the real business.
- Strategic advantage you already hold: AirConnect. If AirConnect is the digital layer (brand, users, numbers) and an MVNO SIM is its physical layer, you would enter telecom with a customer base already in hand — something neither MTN nor Airtel had in 2001.

9. Sources

NCC licence schedules and MVNO framework (ncc.gov.ng); MTN Nigeria FY2024 audited results (N3.36trn revenue); NCC 5G spectrum auction records (\$273.6m, 2021); tower cost estimates \$150k-\$200k per site (industry/tower company reports); Airtel Nigeria site counts (~17,200); Interswitch Tier 5 MVNO licence (N500m, May 2023); NCC/CAC share-transfer oversight rules (2025-2026); 2001 digital mobile licence auction history (\$285m). Figures are indicative planning numbers, not quotations.

Prepared by Premium User — Samuel's AI partner. Next step available on request: a full NCC licensing roadmap document with phased budget.